4. Introduction to Statistical Learning Theory

Nowadays becoming quite common to be working with database of hundreds (or even thousands) of features.

If the no. of features becomes similar (or even bigger!) than the no. of observations stored in a dataset then this can most likely lead to machine learning model suffering from overfitting.

In order to avoid this type of problem, it is necessary to apply either regularization or dimensionality reduction techniques known as feature extraction.

In machine learning, the dimensionale of a dataset is equal to the no. of variables used to represent it.

"Feature Extraction aims to reduce the number of features in a dataset by creating new features from the existing ones (and then discarding the original features)."

These new reduced set of features should then be able to summarize most of the information contained in the original set of features.

Curse of Dimensionality

Multiple dimensions are hard to think in, impossible to visualize, and, due to the exponential growth of the no. of possible values with each dimension, complete enumeration of all subspaces becomes intractable with increasing dimensionality. This problem is known as the curse of dimensionality.

For a given dataset, features and attributes are nothing but dimensions, there can be any no. of features, any no. of attributes in a given dataset That is called dimensions.

Curse of dimensionality refers to non-intuitive properties of data observed when working in high dimensional space.

E.g. Say you dropped a coin on a 100 meter line. How do you find it? Simple, just walk on the line and search.

But what if its 100 x 100 sq. m field? Its already getting tough trying to search a foot-ball ground for a single coin.

Now what if its 100 x 100 x 100 cu.m space?, means football ground now has 30-story height. So this is the Curse of dimensionality

Dimensionality Reduction Methods

To select useful or necessary dimensions / attributes from a large dataset.

There are 2 main methods of dimensionality reduction

1. feature Extraction
2. feature Selection.

1. feature Extraction -

for all given d dimensions
selecting K dimensions
we select d-K dimensions,
and use all the remaining dimensions as clusters K_1, K_2 and so on.

Some other properties of feature extraction are

- Missing values — Remove all the rows
- Low variance — minimum difference
- High correlation — To consider a dataset by removing other.
- PCA (Principal component Analysis) - To find orthogonal transformation.
- Backward feature elimination - To train data model by 'n', 'n-1', 'n-2' ... features. unless untill error rate is minimized.
- forward feature Elimination - $\{f_1\}, \{f_2\}, \{f_3\}$ → To consider features.
- Random forest - works on the basis of target.

Principal Component Analysis

PCA is an unsupervised, non parametric statistical tech. primarily used for dimensionality reduction in machine learning.

High dimensionality means the dataset has a large no. of features. The primary problem associated with high dimensionality in the ML is model overfitting. which reduces the ability to generalize beyond the examples in the training set.

Mr. Richard Bellman described this phenomenon in 1961 as the Curse of dimensionality. where many algorithms that work fine in low dimensions become intractable when the i/p is high dimensional.

PCA makes maximum variability in the dataset more visible by rotating the axes. PCA identifies a list of the principal axes to describe the underlying dataset before ranking them according to the amount of variance captured by each.

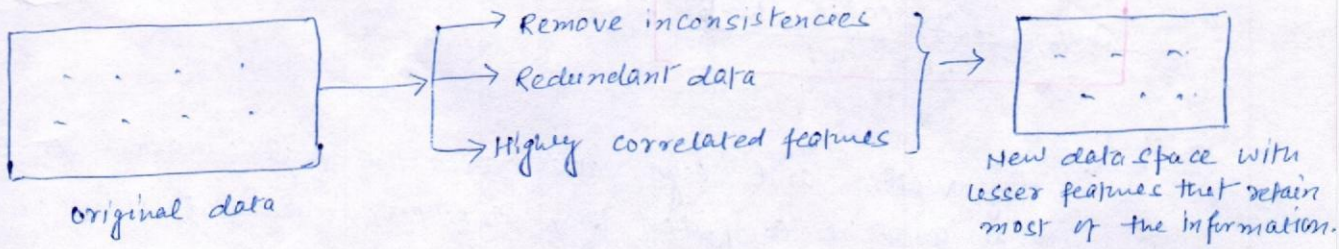
$$PC_1 = W_{1,1}(\text{feature } A) + W_{2,1}(\text{feature } B) + W_{3,1}(\text{feature } C) + \dots + W_{n,1}(\text{feature } N)$$

$$PC_2 = W_{1,2}(\text{feature } A) + W_{2,2}(\text{feature } B) + W_{3,2}(\text{feature } C) + \cdot + W_{n,2}(\text{feature } N)$$

$$PC_3 = W_{1,3}(\text{feature } A) + W_{2,3}(\text{feature } B) + W_{3,3}(\text{feature } C) + \cdot + W_{n,3}(\text{feature } N)$$

The no. of feature combinations is equal to the no. of dimensions of the dataset and in general set the maximum no. of PCAs which can be constructed.

"Principal component analysis is a dim. reduction technique that enables you to identify correlations and patterns in a dataset so that it can be transformed into a dataset of significantly lower dimension without loss of any imp. information".



Steps of PCA

Standardization of the data

↳ computing the covariance matrix

↳ calculating the eigenvectors and eigenvalues

↳ computing the principal components

↳ Reducing the dimensions of the data.

Step 1. Standardization is all about scaling your data in such a way that all the variables and their values lie within a similar range.

$$\boxed{Z = \frac{\text{Variable value} - \text{mean}}{\text{Standard deviation}}}$$

Step 2. A covariance matrix expresses the correlation b/w the different variables in the data set. It is essential to identify heavily dependent variables because they contain biased and redundant information which reduces the overall performance of the model. Covariance matrix for 2 dimensions →

$$C = \begin{bmatrix} Cov(a,a) & Cov(a,b) \\ Cov(b,a) & Cov(b,b) \end{bmatrix} \rightarrow$$

Covariance values denote

How co dependant two variables are

Negative covariance - indirect proportional

Positive covariance - Directly proportional to.

$$\boxed{Cov(a,b) = \sum_{i=1}^{n} \frac{(a_i - \bar{a})(b_i - \bar{b})}{n-1}}$$

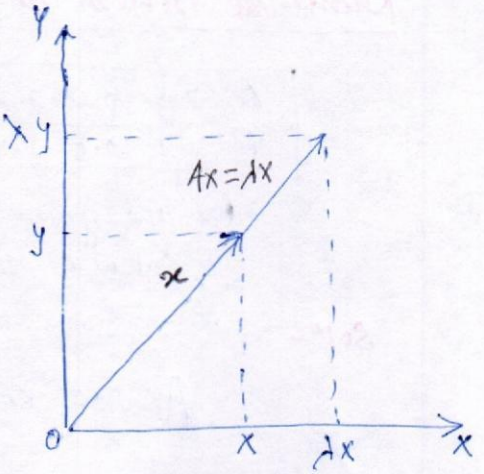
Step-3. Eigenvectors and eigen values are the mathematical constructs that must be computed from the covariance matrix in order to determine the principal components of the dataset.

$$\boxed{C - \lambda I = 0}$$

where I is an identity matrix.

$\lambda \rightarrow$ constant.

Eigenvector describes the angle or directions of the axis through the data space and the
Eigenvalue quantifies the magnitude of the variance of the data on the axis.



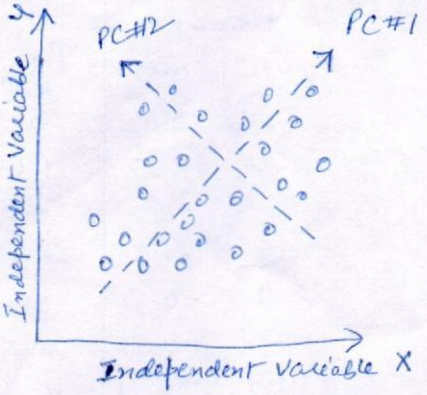
A is an $n \times n$ matrix,
λ is the eigenvalue, and
x is the eigenvector.

⇒ The no. of feature combinations is equal to the no. of dimensions of the dataset.
E.g a dataset with 10 features will have 10 eigenvalues/eigenvector combinations.

⇒ The correlation b/w each principal component should be Zero.
Correlation b/w any pair of eigenvalue/eigenvector is 0 so that the axes are orthogonal, i.e perpendicular to each other in data space.

Step-4

Once we have computed the eigenvectors and eigenvalues, all we have to do is order them in the descending order, where the eigenvector with the highest eigenvalue is the most significant and thus forms the first principal component.



- PC_1 is the most significant and stores the maximum possible information.
- PC_2 is the second most significant PC and stores remaining maximum info and so on.

Step-5

The last step in performing PCA is to re-arrange the original data with the final principal components which represent the maximum and the most significant information of data set.

Numerical Problem on PCA

X	2.5	0.5	2.2	1.9	3.1	2.3	2	1	1.5	1.1
Y	2.4	0.7	2.9	2.2	3.0	2.7	1.6	1.1	1.6	0.9

for the given Two attributes X & Y there are 10 records. Calculate the Principal component analysis.

Sol^n-

As we know that

$$\text{Covariance matrix } C = \begin{bmatrix} Cov(x,x) & Cov(x,y) \\ Cov(y,x) & Cov(y,y) \end{bmatrix}$$

(2x2 matrix because the problem is 2 dimensional)

$$Cov(x,y) = \sum_{i=1}^{n} \frac{(x_i-\bar{x})(y_i-\bar{y})}{n-1} \quad (n=10, \text{given})$$

$\bar{x} = 2.5+0.5+2.2+1.9+3.1+2.3+2+1+1.5+1.1 = 18.1$

$= 18.1/10 = \underline{\underline{1.81}}$

$\bar{y} = 2.4+0.7+2.9+2.2+3.0+2.7+1.6+1.1+1.6+0.9$

$= 19.1/10$

$= \underline{\underline{1.91}}$

for $Cov(x,x) = \sum_{i=1}^{10} \frac{(x_i-\bar{x})(x_i-\bar{x})}{10-1=9}$

X	$X-\bar{X}$	$(X-\bar{X})(X-\bar{X})$
2.5	0.69	0.4761
0.5	-1.31	1.7161
2.2	0.39	0.1521
1.9	0.09	0.0081
3.1	1.29	1.6641
2.3	0.49	0.2401
2	0.19	0.0361
1	-0.81	0.6561
1.5	-0.31	0.0961
1.1	-0.71	0.5041

~~$\Sigma = \frac{6.7387}{9} =$~~

~~$= \underline{\underline{0.7487}}$~~

Sum = 5.5490

now 5.5490/9

$= \underline{\underline{0.6165}}$

for $Cov(y,y) = \sum_{i=1}^{10} \frac{(y_i-\bar{y})(y_i-\bar{y})}{9}$

Y	$Y-\bar{Y}$	$(Y-\bar{Y})(Y-\bar{Y})$
2.4	0.49	0.2401
0.7	-1.21	1.4641
2.9	0.99	0.9801
2.2	0.29	0.0841
3.0	1.09	1.1881
2.7	0.79	0.6241
1.6	-0.31	0.0961
1.1	-0.81	0.6561
1.6	-0.31	0.0961
0.9	-1.01	1.0201

Sum = ~~8.74651/9~~

$= \underline{\underline{0.9718}}$

6.449

6.449/9

$= \underline{\underline{0.7165}}$

for $Cov(x,y)$

x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
2.5	2.4	0.69	0.49	0.3381
0.5	0.7	-1.31	-1.21	1.5851
2.2	2.9	0.39	0.99	0.3861
1.9	2.2	0.09	0.29	0.0261
3.1	3.0	1.29	1.09	1.4061
2.3	2.7	0.49	0.79	0.3871
2.0	1.6	0.19	-0.31	-0.0589
1.0	1.1	-0.81	-0.81	0.6561
1.5	1.6	-0.31	-0.31	0.0961
1.1	0.9	-0.71	-1.01	0.7171

$$Sum = 5.539/9$$
$$= 0.6154$$

So the final matrix is

$$C = \begin{bmatrix} 0.6165 & 0.6154 \\ 0.6154 & 0.7165 \end{bmatrix}$$

Now $C - \lambda I = 0$.

$$\begin{bmatrix} 0.6165 & 0.6154 \\ 0.6154 & 0.7165 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0.6165-\lambda & 0.6154 \\ 0.6154 & 0.7165-\lambda \end{bmatrix} = 0$$

So now find the determinant of the above matrix

$$(0.6165-\lambda)(0.7165-\lambda) - [(0.6154)(0.6154)] = 0$$

$$0.4417 - 0.6165\lambda - 0.7165\lambda + \lambda^2 - 0.3787 = 0$$

$$\lambda^2 - 1.333\lambda + 0.063 = 0$$

$\underline{\lambda_1 = 0.4908}$, $\underline{\lambda_2 = 1.2840}$ → Eigen Values.

Highest Eigen value resultant Eigenvector considered as Principal component of the data set.

for Calculating Eigen vectors

for $\lambda_1 \rightarrow$ $\begin{bmatrix} 0.1257 & 0.6154 \\ 0.6154 & 0.2257 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = 0 \Rightarrow$

$0.1257\, x_1 + 0.6154\, y_1 = 0$ — (1)

$0.6154\, x_1 + 0.2257\, y_1 = 0$ — (2)

Solving (1) & (2)

$x_1 = -0.735$, $y_1 = 0.677$

for λ_2

$$\begin{bmatrix} -0.6675 & 0.6154 \\ 0.6154 & -0.5675 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = 0$$

$$-0.6675\, x_2 + 0.6154\, y_2 = 0 \quad \text{—} (3)$$

$$0.6154\, x_2 + -0.5675\, y_2 = 0 \quad \text{—} (4)$$

By solving Eqn (3) & (4)

$$x_2 = -0.678 \;,\; y_2 = -0.73$$

$$\begin{bmatrix} -0.735 & 0.677 \\ -0.678 & -0.73 \end{bmatrix}$$

→ Principal component.

This eigen vector is according to higher eigen value.

Que. Find the Principal Component for the given 2 dim. data set

x_1	x_2
1.4000	1.6500
1.6000	1.9750
-1.4000	-1.7750
-2.0000	-2.5250
-3.0000	-3.9500
+2.4000	3.0750
1.50000	2.0250
2.3000	2.7500
-3.2000	-4.0500
-4.1000	-4.8500

Que — Analyze the following 3 variate dataset with 10 observations. Each observation consists of 3 measurements on a wafer: thickness, horizontal displacement and vertical displacement.

Clarify the mechanics of Principal Component Analysis

$$X = \begin{vmatrix} 7 & 4 & 3 \\ 4 & 1 & 8 \\ 6 & 3 & 5 \\ 8 & 6 & 1 \\ 8 & 5 & 7 \\ 7 & 2 & 9 \\ 5 & 3 & 3 \\ 9 & 5 & 8 \\ 7 & 4 & 5 \\ 8 & 2 & 2 \end{vmatrix}$$

Feature Selection | Attribute Selection | Variable selection

Problem of selecting some subset of a learning algorithm's input variables upon which it should focus attention, while ignoring the rest.

"Feature selection is one of the core concept in ML which hugely impacts the performance of our model."

Feature selection is the process where we automatically or manually select those features which contribute most to our prediction variable or o/p in which ~~f~~ we are interested in.

Having irrelevant features in our data can decrease the accuracy of the models and make our model learn based on irrelevant features.

Matematically,

- Given a set of features $F = \{f_1, \ldots, f_i \ldots f_n\}$ the feature selection problem is to find a subset that "maximizes the learner's ability to classify patterns"
- formally F' should maximize some scoring function.
- This general definition subsumes feature selection.

Feature Selection can be of 3 types

1. filter methods
2. Wrappers methods
3. Embedded methods

① filter Methods

- Also known as Single factor analysis.
- The predictive power of each individual variable is evaluated.
  - Correlation with the target variable → Target w.r.to x
  - Information value | Information Gain
  - Chi Square Test (Categorical variable)

$y \quad x_1 \; x_2 \ldots$

highest corelated to x is best.

Highest correlation with ~~the~~ target variable

Low correlation b/w predictors → (should not be)

Higher the Information value is the better for the variable.

② Wrapper methods

- uses combination of x variables / feature and find out which one is the best combination.
- Predictive power of the variables is evaluated jointly.
- E.g (x_1, x_2, x_5) are the jointly combination from 10 features.

Set of variables that performs the best

- Subset Selection → fit model with each possible combination of p features
- forward selection
- Backward selection

Total no. of models 2^p, if $p=2$ (lets x_1 & x_2)

$y = B_0 + B_1 x_1$

$y = C_0 + C_1 x_2$

$y = D_0 + D_2 x_2 + D_1 x_1$

4 models as $2^2 = 4$

③ Embedded method

- Also Known as Shrinkage method.
- Inbuilt variable selection methods (without one having to select / reject feature)
- Regularization - controls the value of the parameter. Not so important variables are given very low weight (close to zero)

Subset selection -

Requires massive computational power

To reduce the computational power its broken into 2 stages

e.g $p=3 \Rightarrow 2^3 = 8$

$p=4 \Rightarrow 2^4 = 16$

$p=5 \Rightarrow 2^5 = 32$

but if $p=10$ or more it will more computative.

Stage-1 Fit all combination of models that has only K predictors out of total P predictors. Pick the best model from the set of all K predictors model (lets call this model(K))

Stage 2. Select the one that best from model(1), model(2) ... model(P)

→ Uses RSS, CV error, Adjusted R-square

E.g if $K=1$ $\qquad P=4$

$$\left.\begin{aligned} y &= f(x_1) \\ y &= f(x_2) \\ y &= f(x_3) \\ y &= f(x_4) \end{aligned}\right\} \text{4 models, pick best model}$$

x_1, x_2, x_3, x_4

if $K=2$

$$\left.\begin{aligned} y &= f(x_1, x_2) \\ y &= f(x_1, x_3) \\ y &= f(x_1, x_4) \\ y &= f(x_2, x_3) \\ y &= f(x_3, x_4) \end{aligned}\right\} \text{To select 1 best model.}$$

So for the diff. values of K, for each value of K we need to select best model out of all the models.

<u>Stepwise selection</u> -

$P > 20$ it is almost impossible to use best subset selection Then we apply other approaches like -

⇒ <u>forward stepwise</u>

- Starts with a null model.
- Add predictors to the model one at a time
- Choose the best model among the bests for each K based on RSS.
- if a variable is retained it never drops from the model

	Subset Selection	forward stepwise
1 variable	x_1	x_1
2 variables	$x_1\ x_3$	$x_1\ x_2$
3 variables	$x_1\ x_3\ x_4$	$x_1\ x_2\ x_4$
4 variables	$x_1\ x_2\ x_3\ x_5$	$x_1\ x_2\ x_4\ x_5$

less candidate models for selection — $1 + P(P+1)/2$

Backward Selection :-

it is the reverse of forward : Start with all predictors and then drop at a time and then select the best model.

	Backward stepwise	forward stepwise
	$X_1 X_2 X_3 X_4 X_5$	X_1
	$X_1 X_3 X_4 X_5$	$X_1 X_2$
dropped constraint	$X_1 X_3 X_5$	$X_1 X_2 X_4$
	$X_1 X_5$	$X_1 X_2 X_4 X_5$
	X_1	$X_1 X_2 X_4 X_3 X_5$

E.g $P = 5$

forward → 1, 2, 3 — variable model

Backward → P, p-1, p-2 . . .

- Computational power requirement is similar as that of forward.
- Selection is made through RSS or Deviance.

Hybrid Selection -

- Combines forward and backward selections.
- forward – Variable included never drops off.
- Backward – Variable already dropped from the model never show up in model again

⇒ Hybrid starts with adding one variable at a time like forward but drop variables during the process if a variable no longer improve fit.

- It is more like subset selection – consider more models compared to forward & backward
- It retains the computational advantage of forward & backward selection.

Choosing the best model

Cross validation – K-fold validation

use Adjusted R-square, AIC & BIC